



Curriculum Vitae

Full name: Abdennasser CHEKROUN
Birth date: March, 30 of 1990 in Tlemcen, Algeria
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Institution

Faculty of Technology,
Tlemcen University- Tlemcen - Algeria
And
Laboratory of Mathematics, Analyse Non Linéaire & Mathématiques Appliquées,
Tlemcen University - Tlemcen - Algeria

Professional Activities

Since 2025	Professor of mathematics at Tlemcen university - Tlemcen – Algeria.
2016 - 2025	Lecturer and Research Assistant at Tlemcen university - Tlemcen – Algeria.
2015 - 2016	Temporary Lecturer and Research Assistant (A.T.E.R), Université Claude Bernard Lyon 1, France.
2012–2016	Member of the Mathematics Laboratory, Institut Camille Jordan (ICJ), Claude Bernard Lyon 1, and the Inria-team: Dracula "Multi-Scale Modelling of Cell Dynamics: Application to Hematopoiesis".

Education

2020: University Habilitation, Abou Bekr Belkaid University, Tlemcen, Algeria
2016: Doctorate in mathematics «Contribution to the mathematical analysis of age and space structured partial differential equations describing a cell population dynamics model» Université Claude Bernard Lyon 1, France – Supervisor: Mostafa Adimy
2012: Algerian scholarship training for a PHD in France
2012: Master degree in Dynamical Systems of Population Dynamics and applications, at Abou Bekr Belkaid University, Tlemcen, Algeria
2010: Licence degree in Mathematics (Partial Differential Equation) at Tlemcen University, Algeria
2007: Bachelor in sciences

Research interests

- Delayed differential equations and coupled differential-difference equations
- Reaction-diffusion equations and travelling wave fronts
- Asymptotic stability and Hopf bifurcation
- Transport PDE and modeling in population dynamics (Cell dynamics).
- Blood cell production models (hematopoiesis)

Scientific projects

- Member of Inria project " Modelling Leukemia ", 2012-2015
- Member of Inria project Euromediterranean 3+3 " Mathematical Models and Methods in Cell Dynamics ", 2012-2015
- Member POLONIUM project " Applications of Reaction-Diffusion Equations in Cell " 2013-2015.
- Member of the national project (PRFU) " Modèles mathématiques des phénomènes des sciences de la vie ", 2018-2021.
- Member of the national project (PRFU) " Existence et stabilité des solutions de certaines classes d'équations et d'inclusions différentielles fractionnaires et ses applications ", 2021-2025.

Scientific publications

The research work I have done since my thesis has been the subject of several publications in collaboration with:

Mostafa Adimy, Inria, Université Lyon 1, Institut Camille Jordan, France.
Sabri Bensid, Université de Tlemcen, Algérie.
Soufiane Bentout, Ain Temouchent University, Algérie
Anass Bouchnita, University of Texas at Austin, USA.
Jean-Philippe Berteau, Department of Physical Therapy, City University of New York, USA.
Joydev Chattopadhyay, Indian Statistical Institute, India.
Salih Djilali, Hassiba Benbouali University of Chlef, Algérie.
Charlotte Dugourd-Camus, Inria, Université Lyon 1, Institut Camille Jordan, France.
Abderrahim El Abdllaoui, University of Rabat. Morocco.
Claudia Pio Ferreira, Sao Paulo State University, Brazil
Mohammed Nor Frioui, Laboratoire d'Analyse Nonlinéaire et Mathématiques Appliquées, Université de Tlemcen, Algérie.
Mohamed Helal, Université de Sidi-Bel-Abbes, Algérie.
Bogdan Kazmierczak, Institute of Fundamental Technological Research, Pologne
Toshikazu Kuniya, Graduate School of System Informatics, Kobe University, Japon.
Abdelkader Lakmeche, Université de Sidi-Bel-Abbes, Algérie.
Dibyendu Sekhar Mandal, Amity University Mumbai, India.
Hanene Meghelli, Laboratoire d'Analyse Nonlinéaire et Mathématiques Appliquées, Université de Tlemcen, Algérie.
Mattia Sensi, Politecnico di Torino, Italy.
Laurent Pujo-Menjouet, Institut Camille Jordan, Université Claude Bernard Lyon 1, France.
Gregoire Ranson, , Université Claude Bernard Lyon 1, France.
Tarik Mohammed Touaoula, Laboratoire d'Analyse Nonlinéaire et Mathématiques Appliquées, Université de Tlemcen, Algérie.
Sudip Samanta, Bankura University, India.
Simon Girel, Université Cote d'azur, Nice, France.

Publications

- [1] M. Adimy, A. Chekroun, T. M. Touaoula « A delay defferential-difference system of hematopoietic stem cell dynamics » **C. R. Acad. Sci. Paris, Ser. I**, 353, 303-307, 2015.
- [2] M. Adimy, A. Chekroun, T. M. Touaoula « Age-structured and delay differential-difference model of hematopoietic stem cell dynamics » **Discrete and Continuous Dynamical Systems - Series B**. Volume 20, Number 9, 2765-2791, 2015.
- [3] M. Adimy, A. Chekroun, T. M. Touaoula « Global asymptotic stability for an age-structured model of hematopoietic stem cell dynamics » **Applicable Analysis**, 96 (2016) 429-440.

- [4] M. Adimy, A. Chekroun, B. Kaźmierczak « Traveling wave fronts in a coupled reaction diffusion and difference model of hematopoiesis » **Journal of Differential Equation**, 262 (2017) 4085-4128.
- [5] M. Adimy, A. Chekroun, T. Kuniya « Delayed nonlocal reaction-diffusion model of hematopoietic stem cell dynamics with Dirichlet boundary condition » **MMNP**, 2017.
- [6] A. Chekroun, T. Kuniya « Stability and existence results for a time-delayed nonlocal model of hematopoietic stem cells dynamics » **Journal of Mathematical Analysis and Applications**, 2018.
- [7] A. Chekroun, L. Pujo-Menjouet, J.P. Berteau « A Novel Multiscale Mathematical Model for Building Bone Substitute Materials for Children » **Materials**, 2018.
- [8] A. Chekroun, T. Kuniya « An infection age-space structured SIR epidemic model with Neumann boundary condition » **Applicable Analysis**, 2018.
- [9] A. Chekroun, N. Frioui, T. Kuniya, T. M. Touaoula « Global stability of an age-structured epidemic model with general Lyapunov functional » **Mathematical biosciences and engineering**, 2019.
- [10] M. Adimy, A. Chekroun, T. Kuniya « Coupled reaction-diffusion and difference system of cell cycle dynamics for hematopoiesis process with Dirichlet boundary conditions » **Journal Mathematical Analysis and Applications**, 2019.
- [11] S. Bensid, A. Chekroun, On the multiplicity of solutions of a simplified tumor growth model with a free boundary » **Mathematical Methods in the Applied Sciences**, 2019.
- [12] A. Chekroun, T. Kuniya « An infection age-space-structured SIR epidemic model with Dirichlet boundary condition » **MMNP**, 2019.
- [13] A. Chekroun, Existence and asymptotics of traveling wave fronts for a coupled nonlocal diffusion and difference system with delay, **Electronic Journal of Qualitative Theory of Differential Equations**, 2019.
- [14] M. Adimy, A. Chekroun, C.P. Ferreira ,Global dynamics of a differential-difference system: a case of Kermack-McKendrick SIR model with age-structured protection phase, **Mathematical biosciences and engineering**, 2019.
- [15] A. Chekroun, N. Frioui, T. Kuniya, T. M. Touaoula, Mathematical analysis of an age structured Heroin-cocain epidemic model, **Discrete and Continuous Dynamical Systems - Series B**, 2020.
- [16] A. Chekroun, T. Kuniya, Global threshold dynamics of an infection age-structured SIR epidemic model with diffusion under the Dirichlet boundary condition, **Journal of Differential Equations**, 2020.
- [17] S. Bentout, A. Chekroun, T. Kuniya, Parameter estimation and prediction for coronavirus disease outbreak 2019 (COVID-19) in Algeria, **AIMS Public Health**, 2020.
- [18] K. Nawal, M. Helal, A.Chekroun, A. Lakmeche, Mathematical analysis and global dynamics for a time-delayed Chronic Myeloid Leukemia model with treatment, **Mathematical Modelling of Natural Phenomena**, 2020.

[19] S. Bentout, S. Djilali, A. Chekroun, Global threshold dynamics of an age structured alcoholism model, **International Journal of Biomathematics**, 2021.

[20] M. Adimy, A. Chekroun, T. Kuniya , Traveling waves of a differential-difference diffusive Kermack-McKendrick epidemic model with age-structured protection phase, **Journal of Mathematical Analysis and Application**, 2021.

[21] A. Bouchnita, A. Chekroun, A. Jebrane, Mathematical Modeling Predicts That Strict Social Distancing Measures Would Be Needed to Shorten the Duration of Waves of COVID-19 Infections in Vietnam, **Frontiers in Public Health**, 2021.

[22] D. Mandal, A. Chekroun, S. Samanta, J. Chattopadhyay, A mathematical study of a crop-pest-natural enemy model with Z-type control, **Mathematics and Computers in Simulation**, 2021.

[23] A. Chekroun, L. Pujo-Menjouet, S. Falcoz, K. Tsuen, K. Tsueng , J.P. Berteau, Theoretical Evidence of Osteoblast Self-Inhibition after Activation of the Genetic Regulatory Network Controlling Mineralization, **Journal of Theoretical Biology**, 2022.

[24] M. Adimy, A. Chekroun, T. Kuniya, Global asymptotic stability for a distributed delay differential–difference system of a Kermack–McKendrick SIR model, **Applicable Analysis**, 2022.

[25] M. Adimy, A. Chekroun, B. Kaźmierczak, Traveling waves for reaction-diffusion PDE coupled to difference equation with nonlocal dispersal term and time delay, **Mathematical Modelling of Natural Phenomena**, 2022.

[26] M. Adimy, A. Chekroun, A. EL ABDLLAOUI, A. MARZORATI, Discrete maturity and delay differential-difference model of hematopoietic cell dynamics with applications to acute myelogenous leukemia, **Journal of Biological Systems**, 2022.

[27] A. Chekroun, M. Helal, D. Mandal, A. Lakmeche, M. El Arbi, A Mathematical Study of a Crop-Pest Model With the Caputo Fractional-Order Derivative, **Nonlinear Studies**, 2022.

[28] D. Mandal, S. Samanta, R. Parshad, **A. Chekroun**, M. Helal, J. Chattopadhyay, Study of a Crop-pest-natural Enemy Model with Z-type Control - an Approach to Pest Management, **International Journal of Biomathematics**, 2023.

[29] M. Adimy, A. Chekroun, T. Kuniya, H. Meghilli, Global stability of a SEIR discrete delay differential-difference system with protection phase, **Mathematical Methods in the Applied Sciences**, 2023.

[30] Elise BERTHEL, Laurent PUJO, Eymeric Le Reun, Laurène Sonzogni, Joëlle Al-Choboq, Abdennasser Chekroun, Adeline Granzotto, Clement Devic, Mélanie L Ferlazzo, Sandrine PEREIRA, Michel Bourguignon, Nicolas Foray, Toward an early diagnosis for Alzheimer's disease based on the perinuclear localization of the ATM protein, **Cells**, 2023.

[31] M. Adimy, A. Chekroun, C. Dugourd-Camus, H. Meghilli, Global Asymptotic Stability of a Hybrid Differential–Difference System Describing SIR and SIS Epidemic Models with a Protection Phase and a Nonlinear Force of Infection, **Qualitative Theory of Dynamical Systems**, 2023.

- [32] A. Sha, D.S. Mandal, A. Chekroun, Impact of Prey Refuge in a Diffusive Prey Predator Model with Prey Harvesting, Mutually Interfering Predator and Additional Food for Predator, **International Journal of Applied and Computational Mathematics**, 2023.
- [33] M. Adimy, A. Chekroun, L. Pujo-Menjouet, M. Sensi, A multigroup approach to delayed prion production, **Discrete and Continuous Dynamical Systems – B**, 2023.
- [34] M. Adimy, A. Chekroun, G. Ranson, L. Pujo-Menjouet, Stability Analysis of a New Differential-Difference Model Applied to the Pre-exposure Prophylaxis (PrEP) Effect on the Spread of HIV, **Qualitative Theory of Dynamical Systems**, 2024.
- [35] A. Salhi, A. Chekroun, S. Girel, A SEIR epidemic model with quarantine and distributed delay: revealing the occurrence of recurrent epidemic waves, **Nonlinear Studies**, 2024.
- [36] M. Adimy, A. Chekroun, C. Dugourd-Camus, Is it more effective to protect susceptible individuals or isolate infected people to prevent the spread of a SIR-type epidemic ? **Discrete and Continuous Dynamical Systems – B**, 2024.
- [37] H. Meghilli, A. Chekroun, M. Adimy, Traveling Waves in a Hybrid Reaction-Diffusion-Difference SEIR Epidemic Model With Nonlocal Transmission and Protection Phase With Delay, **Mathematical Methods in the Applied Sciences**, 2025.
- [38] J.P. Berteau, A. Chekroun, L.P. Menjouet, Y-H. Kevin Yang, Stiffness-sensitive gene regulation in human mesenchymal stem cells: Modelling mechanotransduction to predict mineralization and bone protein expression, **Journal of Theoretical Biology**, 2025.
- [39] K. Attar, A. Chekroun, T. M. Touaoula, Traveling waves in a delayed coupled reaction-diffusion-difference system beyond monotonicity, **Discrete and Continuous Dynamical Systems – B**, 2025.
- [40] M. Adimy, A. Chekroun, T. Kuniya, Global dynamics of a differential-difference diffusive SIR model with Dirichlet boundary conditions, **Applicable Analysis**, 2026.
- [41] M. Bouziane, S. Bercisse, A. Chekroun, S. Girel, How delay, isolation and vaccination shape epidemic waves: a bifurcation approach in mathematical epidemiology, **Journal of Applied Mathematics and Computing**, 2026.
- [42] M. Adimy, M. Calero, A. Chekroun, L.P. Menjouet, Travelling waves for delayed reaction–diffusion systems with semi-mixed quasi-monotonicity: a prion disease model, **Proceedings of the Royal Society A**, 2026.

Participation in educational activities through a presentation

- Seminars (with Talk) : 4 (Lyon), 1 (Paris), 4 (Tlemcen), 2 (New York), 1 (Hanoi).
- Conferences (with Talk) : 7 (Algérie), 7 (France), 1 (Maroc), 1 (Espagne), 2 (Italie), 1 (Hongrie), 2 (USA).
- School (with Talk) : 2 (France), 1 (Tunisie).

Participation in Conferences, Workshops and Schools

July 23-30 / 2026 International Congress of Mathematicians ICM2026: « Traveling wave solutions in a delayed hybrid reaction-diffusion and difference SIR epidemic model with protection phase » Philadelphia, USA / **Présentation orale.**

October 19-28 / 2025 EDP et applications: « Existence of traveling wave solutions in a delayed hybrid reaction-diffusion and difference Kermack-McKendrick model with protection phase » Tlemcen, Algeria / **Invited speaker**

December 04 / 2025 National Biomathematical Day: « Nonlocal Coupled Delayed Reaction–Diffusion Systems: Traveling Waves and Propagation Dynamics » Sidi Bel Abbes, Algeria / **Invited speaker.**

November 27-28 / 2024 JSB-2024: « Existence of a traveling wave solution in a Kermack-Mackendrick Model with protection phase and diffusion » Sidi Bel Abbes, Algeria / **Invited speaker.**

October 02-06 2023 GE2MI -2023: « Reaction-diffusion PDE coupled to integral equation with nonlocal dispersal term and time delay» Levico, Italy / **Invited speaker.**

June 25-27 2023 IBMD-2023: « On some infection age-space structured SIR epidemic models with Neumann/Dirichlet boundary conditions » Sidi Bel Abbes, Algeria / **Invited speaker.**

June 02 2023 Bonitos: « Predicting Bone Protein Expression: A Stiffness-Dependent Gene Regulation Model » Lyon, France / **Invited speaker.**

February 8-11 2022 DSABNS 2022 Virtual: « Differential difference Kermack-Mckendrick epidemic model with age-structured protection phase » Spain / **Oral presentation (Virtual).**

June 13-17 2021 Virtual SMB annual meeting: « Traveling waves of a differential-difference diffusive Kermack-McKendrick epidemic model with age-structured protection phase » California / **Oral presentation (Virtual).**

Novembre 18-22 2019 Conference of the Euro-Maghreb International Research: « Coupled reaction-diffusion and difference system with Dirichlet boundary conditions for hematopoiesis process phase » Madrid, Espagne / **Oral presentation.**

Juillet 08-12 2019 DMEE Summer School: « Global dynamics of a differential-difference system: a case of Kermack-McKendrick SIR model with age-structured protection phase » Paris, France / **Oral presentation.**

Février 23-27 Congrès TAM-TAM: « Coupled reaction-diffusion and difference system of 2019 cell cycle dynamics for hematopoiesis process with Dirichlet BC » Tlemcen, Algérie / **Oral presentation.**

Février 07-09 2018 Workshop Dynamical Systems Applied to Biology and Natural Sciences: « Delayed nonlocal reaction-diffusion model for hematopoietic stem cell Dynamics – Dirichlet BC » Torino, Italy / **Oral presentation.**

Mai 01-04 2017 Deuxième Workshop Algéro-français EDP ET Applications: « Contribution to the mathematical analysis of age and space structured partial differential equations

describing a cell population dynamics model » Tlemcen, Algérie / **Oral presentation.**

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| Novembre 08-09
2016 | Congrès des mathématiciens algériens « CMA2016 »
Batna, Algérie / Oral presentation. |
| Juin 01-03
2016 | Perpignan Conference: « EMERGING TRENDS IN APPLIED MATHEMATICS
AND MECHANICS » Perpignan, France / Oral presentation. |
| Mai 23-24
2016 | SIMBAD (Séminaire Itinérant de Mathématiques, Biologie et Applications
des Doctorants) Paris (Pierre et Marie Curie), France / Oral presentation. |
| Juillet 06-10
2015 | Equadiff 2015: « Traveling wave fronts in coupled delayed reaction diffusion
equation and difference equation » Lyon, France / Presentation of a poster.
(second best poster). |
| Juillet 01-04
2015 | 10th Colloquium: « Qualitative Theory of Differential Equations » Szeged,
Hongrie / Oral presentation. |
| Septembre 14-16
2014 | Workshop: « Workshop on mathematics for life sciences»
Sidi Bel Abbès, Algérie / Oral presentation. |
| Juin 16-19
2014 | Poitiers conference: « Conférence Franco-Algérienne EDP et applications »
Poitiers, France / Oral presentation. |
| Janvier 27-28
2014 | Workshop under EuroMediterranean program : « Modèles et Méthodes
Mathématiques en Dynamique Cellulaire » Marrakech, Maroc / Oral
presentation. |
| Décembre 03-05
2013 | UABB Tlemcen university school researchers: « Modélisation des Systèmes
Complexes » Tlemcen, Algérie / Oral presentation. |
| Juin 10-13,
2013 | USTHB University: « Conférence sur les Systèmes Dynamiques Complexes »
Alger, Algérie / Oral presentation. |
| Mai 27-31
2013 | ESMTB-EMS Summer school : « Multiscale modeling in the life sciences »
Lyon, France / Presentation of a poster. |

Organization of Scientific Events

- Participated in the International Biomathematical Modeling Days (IBMD-2023), held in Sidi Bel Abbès, Algeria, in 2023, as both an attendee and organizer.
- Participated in the 9th TAMTAM Conference, held in Tlemcen, Algeria, in 2019, as both an attendee and organizer.
- Participated in the 2nd Algerian-French Workshop on PDEs and Applications, held in Tlemcen, Algeria, in 2017, as both an attendee and organizer.

Scientific research visit

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| November 17-21
2014 | Institute of fundamental technological research (Warsaw) Polish academy of
sciences, France-Polonium project: « Applications of reaction-diffusion
equations in biology and medicine » Warsaw, Poland. Oral presentation. |
| October 26-31
2015 | Institute of fundamental technological research (Warsaw) Polish academy of
sciences, France-Polonium project: Warsaw, Poland. |

December 2021 Institut Camille Jordan: Audin chair from Insmi 2020

Distinction

- Selected and support to attend ICM 2026 in Philadelphia, with a short communication.
- Awarded the IMU–Simons Research Fellowship, 2024
- First winner of the Audin chair from CNRS 2020, I was welcomed for 1 month at the Camille Jordan Institute for work in collaboration with Laurent Pujo-Menjouet, lecturer at Claude Bernard University.
- Second best poster presentation: Equadiff 2015 (July 06-10) “Traveling wave fronts in coupled delayed reaction diffusion equation and difference equation” Lyon, France.
- Obtaining an excellence scholarship (PROFAS A) from the Algerian ministry for doctoral training in France 2012-2016).

Academic services: Review for scientific journals Chapter Springer, Journal of Dynamics and Differential Equations, International Journal of Biomathematics, Mathematical Modelling of Natural Phenomena, Nonlinear Analysis: Real World Applications, Qualitative theory of dynamical systems.

Teaching

2015-2016	Lyon 1 university, Analysis L1, Portail Math TD 36h. Lyon 1 university, Analysis L1, Portail Info TD 36h. Lyon 1 university, Analysis L1, Portail Cours préparatoires Khôlle 24h.
2016-2025	Tlemcen university, Analysis L2, Portail Civil engineering Courses and TD. Tlemcen university, Prob-stat L2, Portail Civil engineering Courses and TD. Tlemcen university, Complex analysis L2, Portail Civil engineering Courses and TD.
2024-2025	Analysis L2, Algebra L1, Portail Civil engineering.
2025-2026	Analysis L1, Analysis L2, Portail Civil engineering.

Teaching production

- Handout 1: Statistiques descriptives et exercices (Rappels de cours et exercices corrigés sur la statistique descriptive), 2017.
- Handout 2: Exercices corrigés pour l’analyse complexe, 2020.
- Book: A. Chekroun, « Statistiques descriptives et exercices : Rappels de cours et exercices corrigés sur la statistique descriptive » Generis Publishing, 2021.

Supervision and juries

- 6 Master's theses at the University of Tlemcen.
- 1 PhD, 1 PhD (in progress), 1 doctoral co-supervision (in progress)
- Examination: 4 doctoral jury.

Skills

Data processing: office, Maple, C++, Matlab, Latex and other tools.